

Analyzing Chaos from Phase-Space Reconstruction

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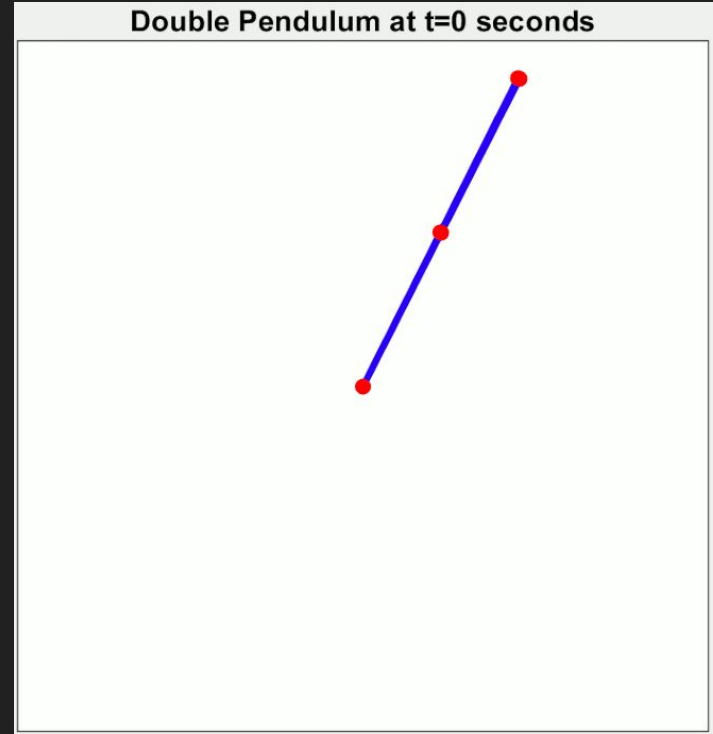
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What is Chaos?

Two defining characteristics:

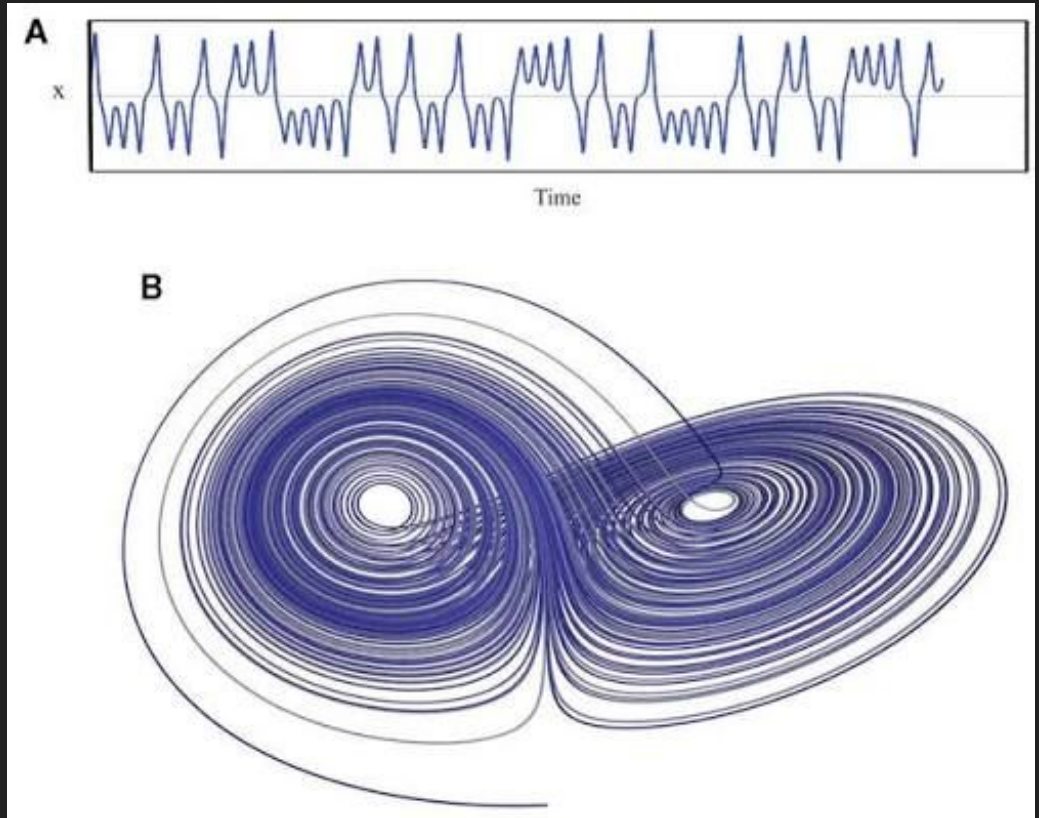
- Deterministic: The system follows precise rules, and two systems with the same initial conditions evolve into the same states.
- Sensitive: Changes in the initial condition result in large exponential errors over time.

Important for understanding the predictability of such systems, which frequently appear in nature through hurricanes, tornadoes, and fluid dynamics.



The Objective

In this project we would like to develop a robust method of detecting chaos in a physical system from data streams, using a benchmark from the proven chaotic Lorenz system.



Source:

https://www.researchgate.net/publication/51855958_Heart_Rate_Variability_-_A_Historical_Perspective

The Method

Takens' Theorem: a reconstruction of chaotic dynamics using a sufficient embedding dimension and time delay will have similar geometric properties.

Step 1

Calculate the Average Mutual Information (AMI) and find τ (tau).

Step 2

Determine the embedding dimension of the system.

Step 3

Calculate the maximum Lyapunov exponent and reconstruct a similar manifold.

Step 4

Compare exponent to surrogate data, which ensures it is a strong sign of chaos.

Step 1: AMI

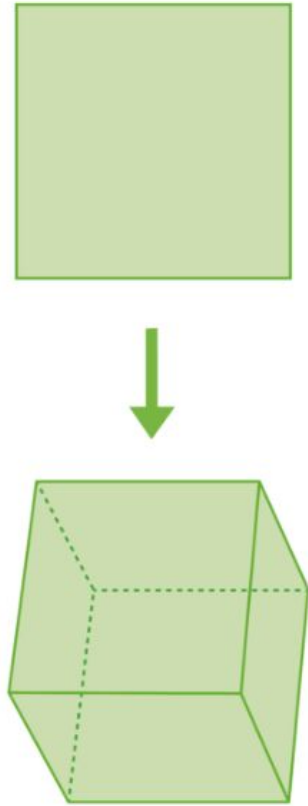
- Average Mutual Information in our context measures how well a point τ time units into the future corresponds to the original point.
- It works by taking the original series, creating a new series with each index shifted τ to the right, and comparing how often values from both series fall into specific ranges of data called bins.
- These probabilities are then plugged into this formula to calculate the AMI of a specific time lag τ :

$$I(\tau) = \sum_{x_t} \sum_{x_{t+\tau}} p(x_t, x_{t+\tau}) \log_2 \left(\frac{p(x_t, x_{t+\tau})}{p(x_t)p(x_{t+\tau})} \right)$$

- We are mainly concerned with the first local minimum on the graph of AMI values vs. τ , as this represents the first value where the correlation between the points is reduced sufficiently.

Step 2: Embedding Dimension

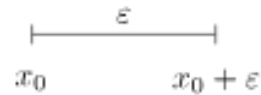
- Given a single set of data values for a system, we need to calculate its dimension in order to reconstruct it, called the embedding dimension.
- Done by doing the False Nearest Neighbors algorithm, which for a dimension m checks the plot of $[x_t, x_{t+\tau}, \dots, x_{t+(m-1)\tau}]$ for the number points with close Euclidean distance until plotted in the next dimension.
- We run this algorithm until the percentage of “false neighbors” stabilizes close to 0; the dimension it settles around will be our m -value.



Step 3: Maximum Lyapunov Exponent

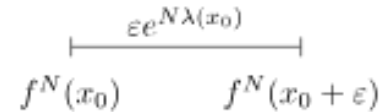
- The Lyapunov Exponent of a system measures how quickly slight perturbations in initial conditions cause divergence in future values, and positive Lyapunov exponents are a strong indicator of chaos.
- We use the Rosenstein method: locate the Nearest Neighbor of each point, track the distance between the points over a certain amount of time steps, average the logarithms of each pair, and plot these values.
- The slope before the graph starts to plateau represents our Maximum Lyapunov Exponent.

Distance between two start values:



A horizontal line segment with vertical tick marks at each end. Above the segment is the Greek letter epsilon (ϵ). Below the left tick mark is the label x_0 , and below the right tick mark is the label $x_0 + \epsilon$.

After N iterations:



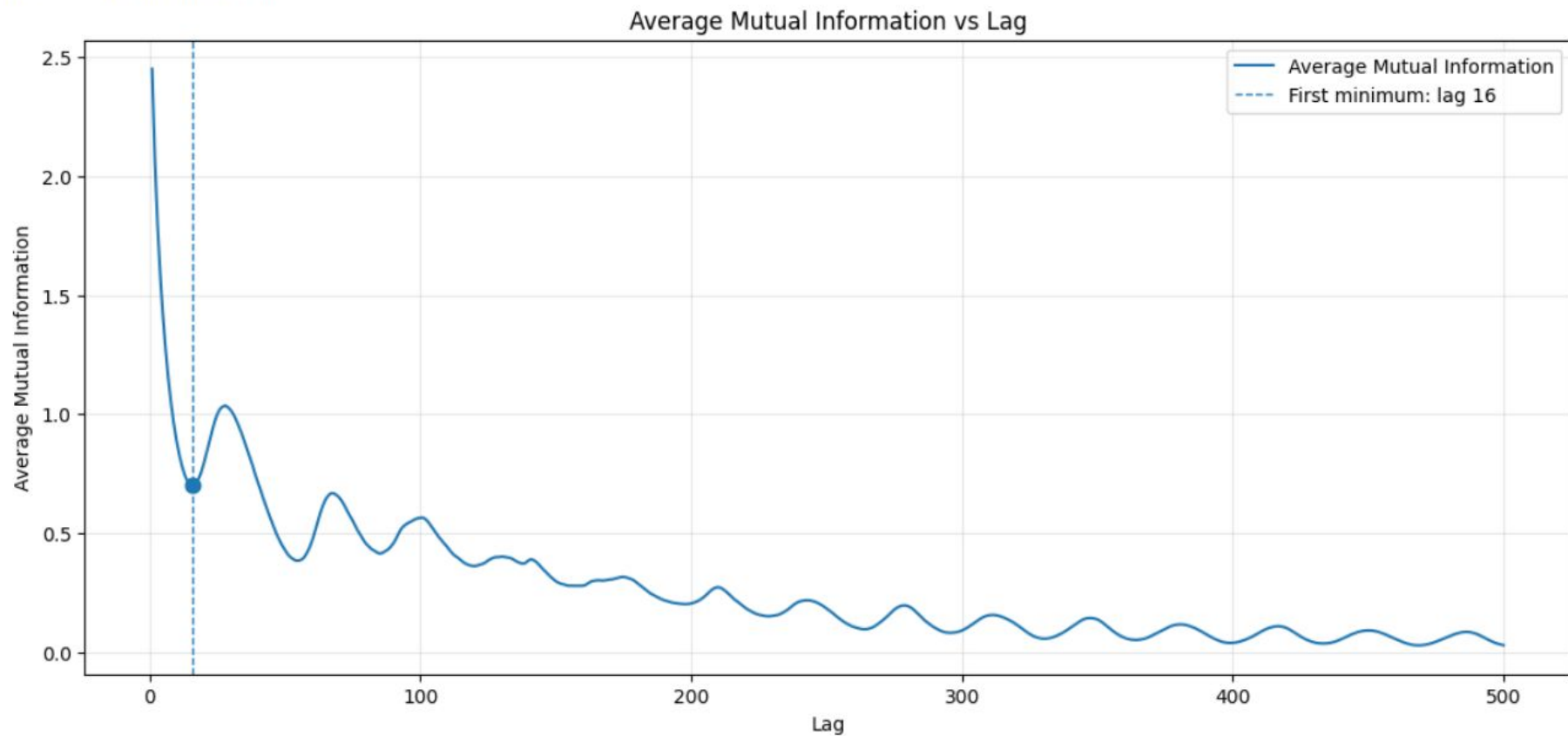
A horizontal line segment with vertical tick marks at each end. Above the segment is the expression $\epsilon e^{N\lambda(x_0)}$. Below the left tick mark is the label $f^N(x_0)$, and below the right tick mark is the label $f^N(x_0 + \epsilon)$.

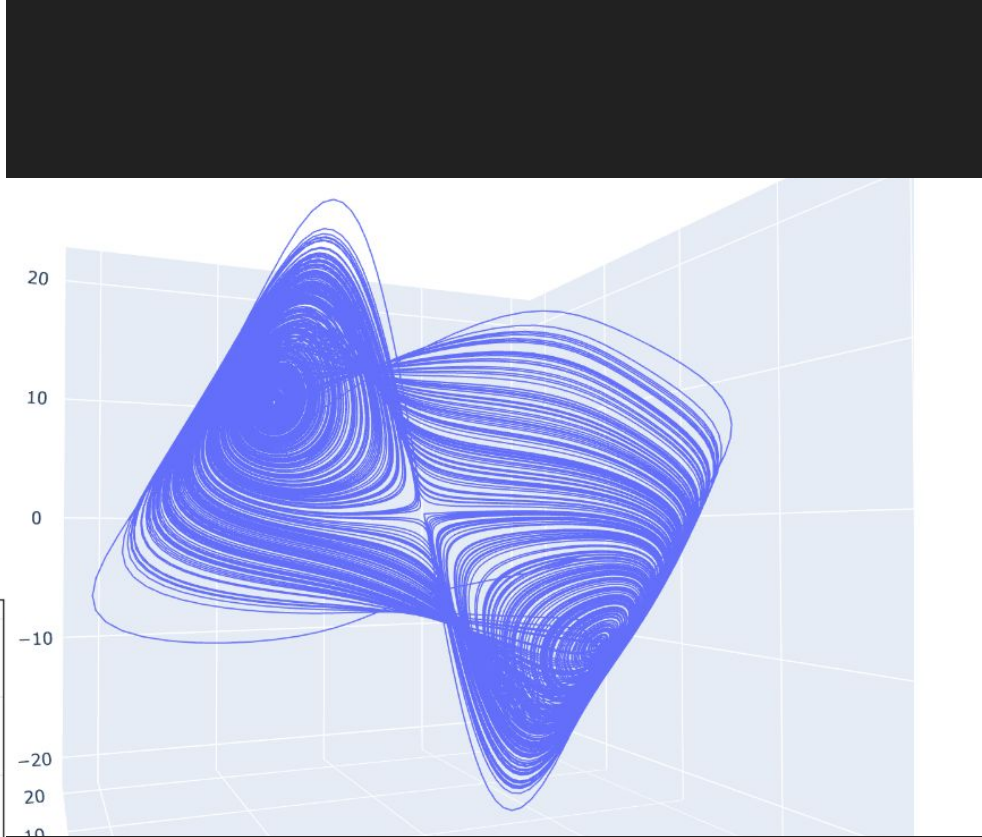
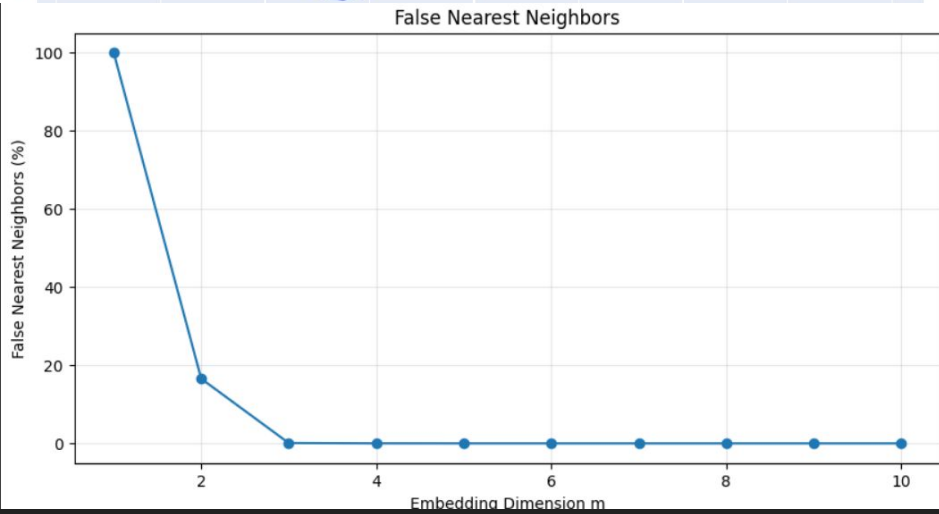
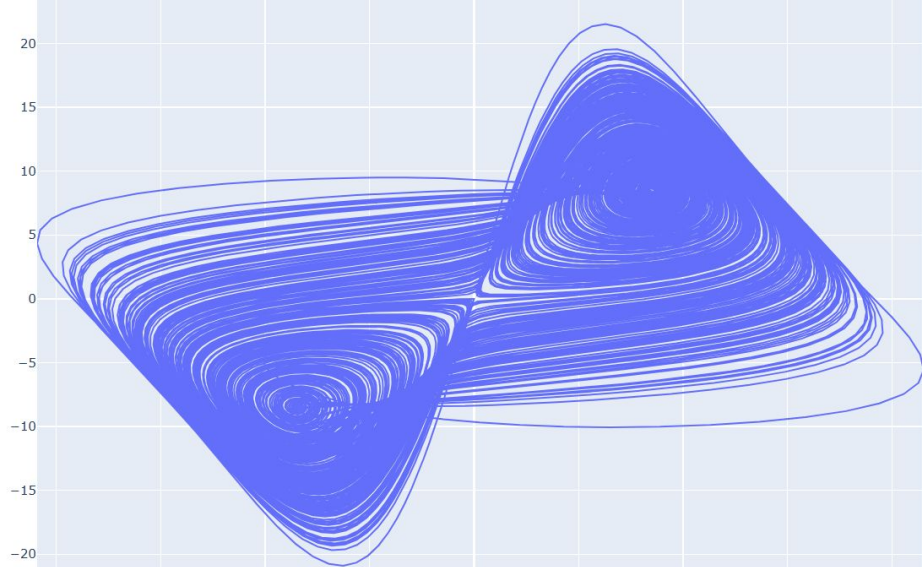
Source: https://en.wikipedia.org/wiki/Lyapunov_exponent

Step 4: Statistical Confidence Check

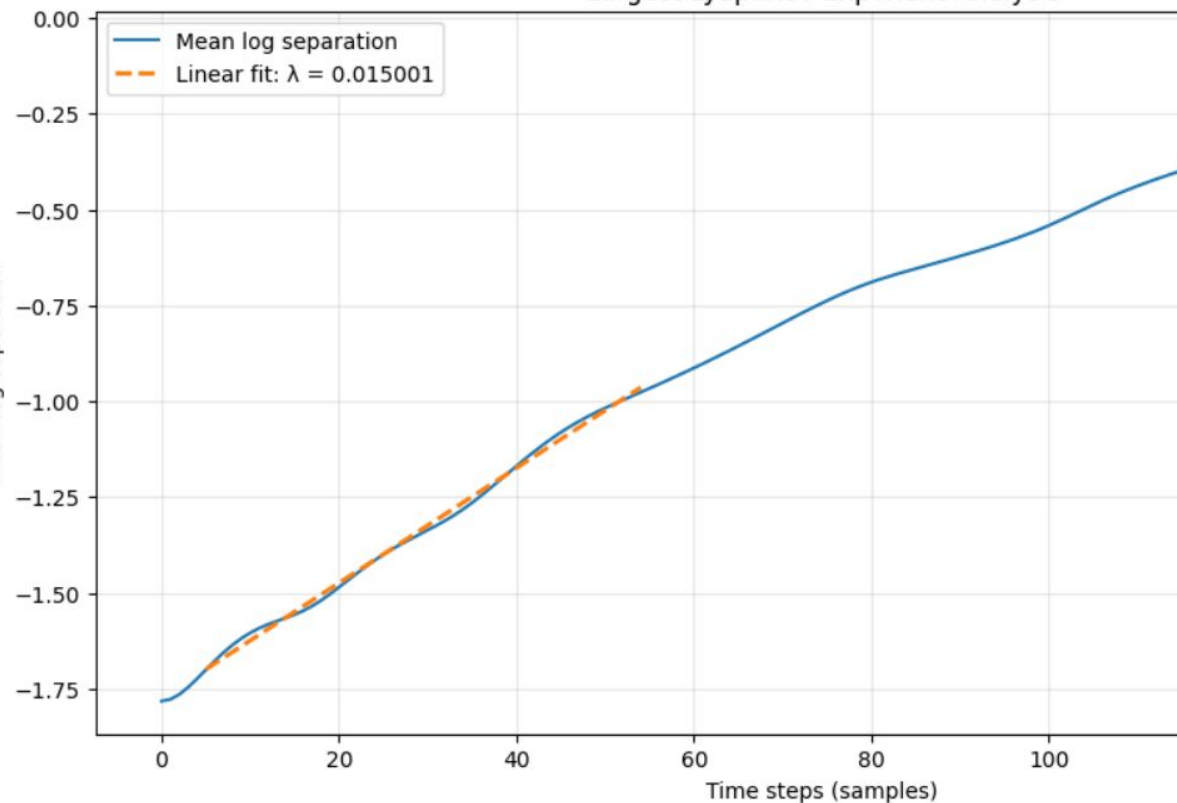
- The key point is that a positive Lyapunov exponent is only an indicator of chaos, though we can be fairly confident.
- However, correlated random data, called stochastic noise, can also produce a positive exponent, so we generate a series of surrogate data and compare their exponents to check if the original's was significantly larger.
- We use Phase Randomized Fourier Surrogates, which preserves all the “peaks” of the data but randomizes their frequencies and where they appear, essentially creating duplicates that share certain properties but have an underlying true randomness to them.

First local minimum:
Lag = 16
AMI = 0.7018640146358154





Largest Lyapunov Exponent Analysis



Original prediction score: 0.035185

Mean surrogate score: 0.529237

Median surrogate score: 0.528760

Minimum surrogate score: 0.513641

Maximum surrogate score: 0.549787

Mean surrogate NRMSE by horizon:

Horizon 1: 0.067193

Horizon 5: 0.229419

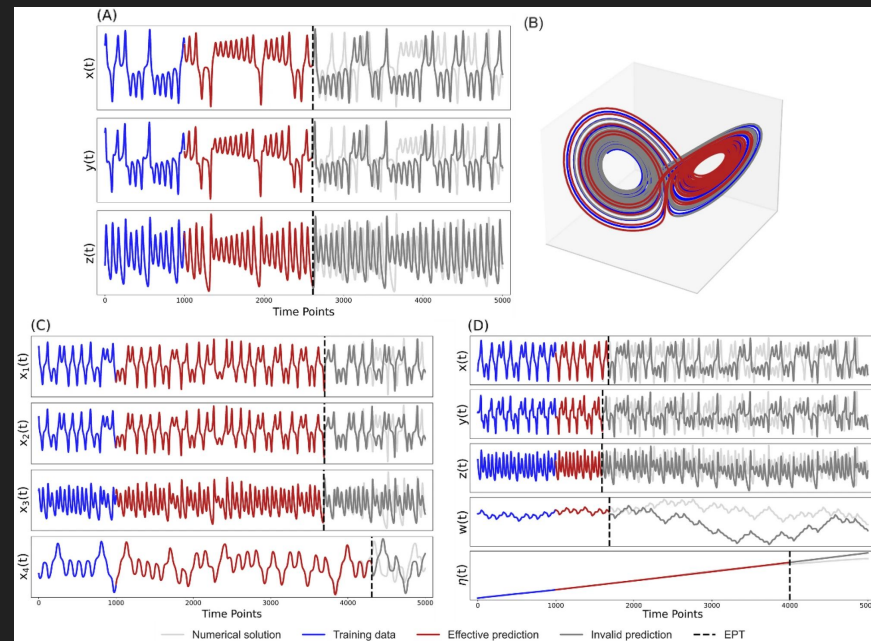
Horizon 10: 0.464398

Horizon 20: 0.823640

Horizon 40: 1.061535

So What?

- Our method was able to pass the benchmark and suggest the presence of chaos in a known chaotic system! We were also able to reconstruct a model of the attractor.
- Despite these conclusions, being confident does not prove chaos, but it is sufficient for prediction models.



Source: <https://rdcu.be/qhcB5RKTYxYa>